GROMM

Generative Risk-neutral Option Market Models

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QuantMinds International 2026

Agenda

1. **Why:** desks need dynamics, not just today's fit
2. **The model:** local volatility on a grid, with an explicit risk-neutral drift
3. **Evidence:** S&P 500, and the same arbitrage test against other generators
4. **Outlook:** $\mathbb{P}$ and $\mathbb{Q}$ in one model, hedging

Why

desks need dynamics, not just today's fit

What desks run today

Local-stochastic volatility (LSV)

$$\sigma^2_{\text{Dupire}}(T,K) = L(T,K)^2\,\mathbb{E}\big[V_T \mid S_T = K\big].$$

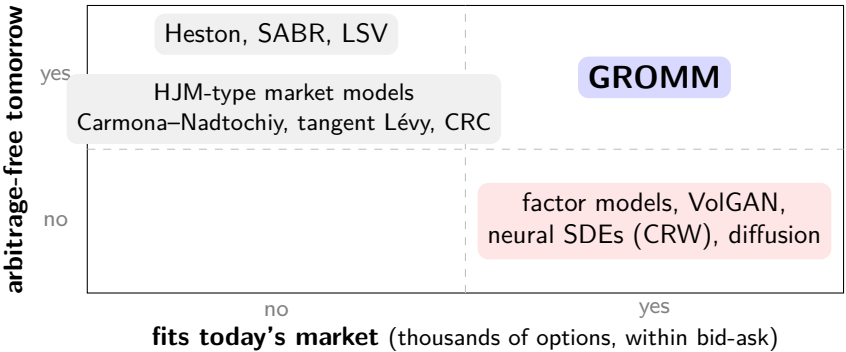
- refitted every day $\Rightarrow$ modelling inconsistency
- SV part calibrated, not estimated from how the surface moves

GROMM

The whole surface is the state: its dynamics are **estimated from history** and **arbitrage-free from one day to the next**.

Dupire (1994). Pricing with a smile; Guyon, Henry-Labordère (2011). The smile calibration problem solved.

The missing model

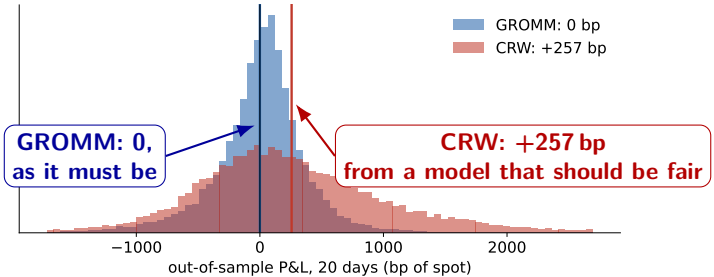


Fits today's market and is arbitrage-free tomorrow, for every option at once.

Schematic.

Why arbitrage-free dynamics matter

Train any optimiser on a simulator, and it finds the simulator's arbitrage:



Out-of-sample P&L of a buy-or-sell strategy, rebalanced daily: 39 options, 20 days.

- Any optimiser that trades the simulated options will find their arbitrage.
- Costs hide it only if drift $<$ spread: no guarantee for a generic simulator.
- **Cleaner: a simulator with an analytic $\mathbb{Q}$.**

The model

local volatility on a grid, with an explicit drift

The state: local volatility on a grid

Fixed cash strikes $K^0 < \cdots < K^{N+1}$, fixed expiries $T_1 < \cdots < T_M$, call prices $C^{j,i} = C(K^i, T_j)$.

$$\Sigma^{j,i} = \frac{C^{j,i} - C^{j-1,i}}{(T_j - T_{j-1})\,\frac{1}{2}\,(K^i)^2\,\Gamma^{j,i}} = \frac{\text{calendar spread}}{\text{butterfly}}$$

discrete Dupire formula, Γ: discrete gamma (second difference in strike)

Key fact (Buehler–Ryskin)

$$\Sigma^{j,i} > 0 \text{ at every node} \quad \Longleftrightarrow \quad \text{no static arbitrage.}$$

So we model $\log \Sigma$: **any value is allowed**.

Buehler, H., Ryskin, E. (2015). Discrete local volatility for large time steps.

The density recursion

q^j: risk-neutral density of S_{T_j} on the grid = butterfly prices.

$$\boxed{q^{j-1} = M^j\, q^j, \qquad M^j = I - \Omega_j \,\mathrm{diag}(\Sigma^j)}$$

one implicit finite-difference step per expiry; Ω_j: fixed tridiagonal matrix

$$q^1 \xrightarrow{\Sigma^2} q^2 \xrightarrow{\Sigma^3} q^3 \longrightarrow \cdots \xrightarrow{\Sigma^M} q^M$$

Front density q^1 + any $\Sigma \geq 0 \implies$ every density, every price.

Itô on the recursion

Input: the volatility of local volatility, any number of factors. **Unknown:** the drift.

$$\frac{\mathrm{d}\Sigma_t^j}{\Sigma_t^j} = \mu_t^j \,\mathrm{d}t + \xi_t^j \,\mathrm{d}W_t.$$

No arbitrage: densities are butterflies, so they are martingales, $\mathrm{d}q_t^j = E_t^j \,\mathrm{d}W_t$.

$$\mathrm{d}q^{j-1} = M^j \,\mathrm{d}q^j + (\mathrm{d}M^j)\, q^j + \mathrm{d}[M^j, q^j]$$

match $\mathrm{d}W$

$$E^j = (M^j)^{-1}(E^{j-1} + \Omega_j B^j q^j)$$

one tridiagonal solve

set $\mathrm{d}t$ **to zero**

$$\mu^{j,i} = -\langle \xi^{j,i}, \eta^{j,i} \rangle, \quad \eta = \frac{E}{q}$$

node by node

And the model exists

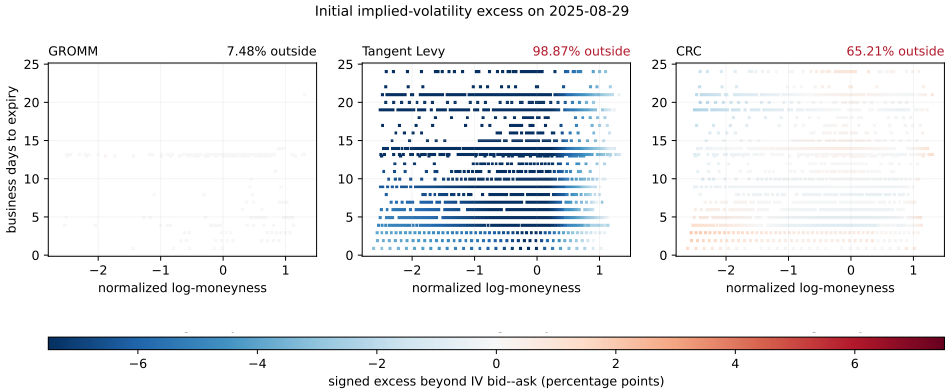
Unique strong solution, any number of factors, no conditions on the wings.

Drift restriction on a fixed lattice: Wissel (2007); continuum: Carmona, Nadtochiy (2009).

Evidence

S&P 500, and the same test against other generators

Day 0: fitting SPX



97.3% of 2.2M SPX quotes (2022–25) inside bid-ask

How to test for dynamic arbitrage: the BHW test

Under $\mathbb{Q}$ every traded price is a martingale:

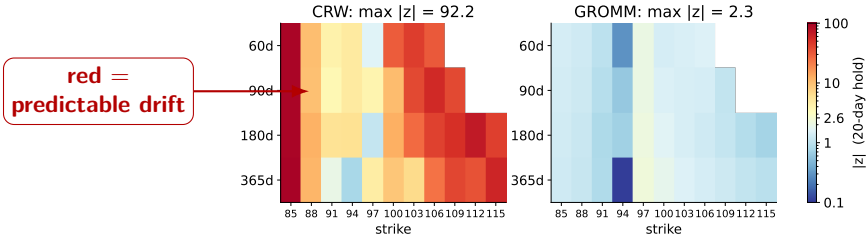
$$\mathbb{E}[C_{t+1} - C_t] = 0 \qquad \text{for every contract and every day.}$$

Simulate many paths and average:

$$z = \frac{\text{mean price change}}{\text{standard error}} \;\sim\; N(0,1) \quad \text{if no arbitrage.}$$

- 1% of tests beyond 2.576 and **none beyond 5**: e.g. 780 tests (39 options $\times$ 20 days) $\Rightarrow$ about 8 beyond 2.576, 0.0005 expected beyond 5
- conditional version: train a trading strategy on half the paths, test it on the other half

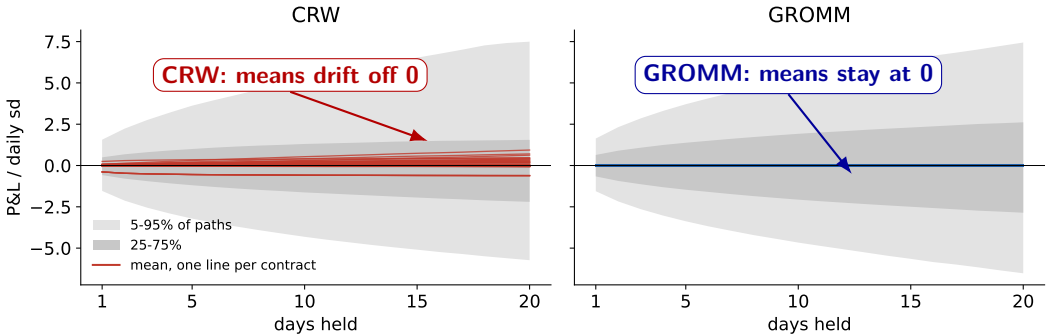
Same data, same BHW test



CRW's own Heston-SLV benchmark and trained models (Cohen, Reisinger, Wang 2023); GROMM fitted to the same data, rank 3 = CRW's 3 Brownian motions; same start state, 39 contracts, 20 days, 48,000 path pairs.

	max $\lvert z \rvert$	tests beyond 5	of
CRW	294	320	724
GROMM	2.4	0	780
exact Heston (control)	3.0	0	723

Same data, same BHW test: every contract



P&L of holding one contract, in units of its daily standard deviation. Grey: spread of individual paths (5–95%, 25–75%). Lines: mean over all paths, one per contract.

SPX, the full grid

5,001 instruments **62,569 BHW tests**

	expected	GROMM
tests beyond 2.576	626	635
tests beyond 5	0	0
conditional T_{sqr}, T_{ent}	$N(0,1)$	0.84, 0.84

5,001 instruments = 25 daily expiries × 200 strikes + spot. 62,569 tests = (instrument, day) pairs that are live and near the money on enough of the 100,000 paths.

Can the test detect arbitrage?

	max $\lvert z \rvert$	beyond 5	of
GROMM under $\mathbb{Q}$	2.7–3.7	0	570
GROMM under $\mathbb{P}$	342	570	570
generalised Bates (CRC) under $\mathbb{Q}$	140	320	570

- $\mathbb{P}$ carries risk premia, so it *should* fail: the test detects drift.
- The same test passes GROMM under $\mathbb{Q}$.

Outlook

$\mathbb{P}$ and $\mathbb{Q}$ in one model, hedging

$\mathbb{P}$ and $\mathbb{Q}$ in one model

$$\underbrace{\mu^{\mathbb{Q}} = -\langle \xi, \eta \rangle}_{\text{computed}} \qquad \mu^{\mathbb{P}} = \mu^{\mathbb{Q}} + \xi\, \theta_t$$

- same state, same loadings ξ, estimated from history
- θ_t: market price of risk. AR(1) in the paper; any model can be used
- simulate under $\mathbb{P}$ for risk, under $\mathbb{Q}$ for pricing and hedging

Outlook: hedging

Bellman hedging needs one expectation per step. GROMM gives its generator in closed form:

$$\mathbb{E}[V(s_{t+\Delta}) \mid s_t] = V + \Delta\Big(\mu \cdot \nabla V + \tfrac{1}{2} \sum_{f=1}^{k} \xi_f^\top \nabla^2 V \, \xi_f\Big) + O(\Delta^2).$$

- small steps: a PDE with **known, arbitrage-free coefficients** for the entire surface
- no nested Monte Carlo; k Hessian–vector products per step
- matters with trading costs and a limited set of hedges, where replication is no longer optimal

What GROMM does, and what it doesn't (yet)

Does

- explicit $\mathbb{Q}$ for every option on the grid
- no static and no dynamic arbitrage, by construction
- fits S&P 500 within bid-ask, including the next-day expiry
- $\mathbb{P}$ and $\mathbb{Q}$ from one state
- fast: one tridiagonal solve per factor and expiry; 96,000 paths $\times$ 20 days in about 4 minutes on one CPU core

Doesn't (yet)

- loadings estimated from history: the number of factors is limited by the data
- continuous paths: no jumps

Takeaways

1. **State = local volatility on a grid:** static arbitrage is impossible by construction.
2. **Itô on the density recursion:** an explicit drift, node by node. Dynamic arbitrage is impossible by construction.
3. **On SPX:** 97% inside bid-ask, and 62,569 BHW arbitrage tests passed. Neural competitors fail the same test.

arXiv:2610.16732

Please ask questions

arXiv:2610.16732

Appendix: who has tried (1/2), generative and ML simulators

	model	no static arb.	explicit $\mathbb{Q}$	grid (expiries $\times$ strikes)
Wiese, Bai, Wood, Buehler (2019)	DLV, PCA and GAN	✓ on grid	×	8×4
Wiese et al. (2021)	DLV latent flow	✓ on grid	×	9×4 per asset
Medvedev, Wang (2022)	LSTM / ConvLSTM on IVs	×	×	
Cohen, Reisinger, Wang (2023)	neural SDE	✓ on grid	∼ penalty	5×11
Cont, Vuletić (2023)	IV factors, path reweighting	∼	×	fixed IV grid
Zhang, Li, Zhang (2023)	VAE and neural surface	∼	×	14×11
Ning, Jaimungal, Zhang, Bergeron (2023)	VAE to pricing-model parameters	✓	×	parametric
Choudhary, Jaimungal, Bergeron (2024)	functional PCA, neural SDE	×	×	17×11
Vuletić, Cont (2024)	VolGAN	∼	×	
Jin, Agarwal (2025)	conditional diffusion	∼	×	9×9
Brooks et al. (2026)	penalised VAE, latent flow	∼	×	32×16
Hao, Ji (2026)	diffusion with refinement	∼	×	15×17
Han et al. (2026)	sequential diffusion	∼	×	11×9
Yang, Wang, Zhang, Li (2026)	diffusion across equities	∼	×	11×9 per stock
François et al. (2026)	five-factor IV with SPX	∼	×	
Buchegger, Gonon (2026)	latent diffusion	∼	×	
GROMM	**stochastic DLV**	✓	✓	$\mathbf{25 \times 200}$

∼: approximate (penalties, reweighting, measured violation rates). Full classification: paper, Appendix.

Appendix: who has tried (2/2), market models

	model	no static arb.	explicit $\mathbb{Q}$	fit / scale
Cont, da Fonseca (2002); Fengler, Härdle, Villa (2003)	IV factor models	×	×	empirical $\mathbb{P}$
Schweizer, Wissel (2008)	local implied volatility	✓	✓ theory	one expiry
Carmona, Nadtochiy (2009)	local-volatility codebook	—	drift only	no existence proof
Carmona, Ma, Nadtochiy (2017)	tangent Lévy	✓	✓	98.9% outside bid-ask*
Gambara, Teichmann (2021)	consistent recalibration (Bates)	✓	∼ fails as calibrated	65.2% outside bid-ask*
Buehler, Horvath, Kratsios, Wiese (2026)	DySANOS	✓	∼ 48 options	SPX
GROMM	**stochastic DLV**	✓	✓ **5,000 options**	**97.3% inside bid-ask**

*our implementation, SPX on 29 Aug 2025.